C6.7 Interoperability considerations

A system wide coherency protocol has to work correctly with components that might have:

- different structures for caching and storing data
- different cache line sizes
- different physical address space sizes.

C6.7.1 Cache Line size conversions

Maximum performance and efficiency is usually achieved when all components use the same cache line size. For systems where this is not possible, it is the responsibility of the interconnect to convert between the different cache line sizes.

Note

The supported cache line sizes and maximum physical address space size are defined at design time.

Narrow to wide conversion

When the master initiating a transaction has a narrow cache line, the following conversion is required:

- A read transaction can be converted to a wider cache line size. The transactions that can be converted are:
  - ReadOnce
  - ReadClean
  - ReadNotSharedDirty
  - ReadShared
  - ReadUnique.

  The converted transaction fetches the data that is required to complete the original transaction together with data that is not required. The excess data must be written back to main memory if it is dirty, but can be discarded if it is clean.
- A clean transaction can be converted to a wider cache line size. The transactions that can be converted are:
  - CleanUnique
  - CleanShared
  - CleanInvalid.

  Dirty data obtained as a result of the clean transaction must be written back to main memory.
- The MakeUnique or MakeInvalid transaction require special consideration. A cache line that is wider than that requested by a master with a narrow cache line cannot be invalidated. When converting a MakeUnique or MakeInvalid transaction to a wider cache line size, it must be converted to a CleanInvalid transaction. This ensures that all dirty data is written back to main memory before the wider line is invalidated.

  Note

  Similar consideration is required for the WriteUnique or WriteLineUnique transaction.

Wide to narrow conversion

When the master initiating a transaction has a wide cache line, the transaction can be broken into multiple narrow transactions.

Each of these narrow transactions can be responded to by different cached masters during the snoop process and some of the narrow transactions might require access to main memory.

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It is the responsibility of the interconnect to:

- assemble the transaction response sent to the originating master
- ensure that the multiple narrow transactions are sequenced correctly, that is, as a contiguous block with respect to other snoop transactions.

If any part of the wide cache line is shared, then the whole cache line must be considered as shared. If any part of the wide cache line is dirty, then the whole cache line must be considered as dirty.

The passing of dirty data from a snooped master is optional for the following transactions:

- ReadOnce
- ReadClean
- ReadNotSharedDirty
- ReadShared.

It is the responsibility of the interconnect to ensure that no parts of the cache line can be dirty in more than one cache.

C6.7.2 Additional Cache Line conversion considerations

The following transactions, issued by a master, are not required to be a full cache line size:

- ReadNoSnoop
- ReadOnce
- WriteNoSnoop
- WriteUnique
- WriteBack
- WriteClean.

All other transactions are required to be a full cache line size and must use the full width of the data bus.

As a snoop transaction is required to be a full cache line size, it is the responsibility of the interconnect to carry out the required size translation. Size translation is required for:

- a ReadOnce transaction that is converted into a ReadOnce snoop
- a WriteUnique transaction that is converted into a CleanInvalid snoop.

C6.7.3 Address space size

The protocol supports communication between components that have different physical address space sizes.

Components with different physical address space sizes must communicate as follows:

- The component with the smaller physical address space must be positioned within an aligned window in the larger physical address space. Typically, the window is located at the bottom of the larger physical address space. However, it is acceptable for the component with the smaller physical address space to be positioned in an offset window within the larger physical address space.
- An outgoing transaction must have the required additional higher-order bits added to the transaction address.
- An incoming transaction must be examined so that:
  - a transaction that is within the address window has the higher-order address bits removed and is passed through
  - a transaction that does not have the required higher-order address bits is suppressed.

Note

It is the responsibility of the interconnect to provide the required functionality.

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Chapter C7
Cache Maintenance

This chapter describes the cache maintenance operations that make loads and stores to specific caches visible to non-coherent agents in the system. It contains the following sections:

- *ARCACHE and ARDOMAIN requirements* on page C7-242
- *Other cache maintenance considerations* on page C7-243.

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